Zhandos Brown

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Education

Boston University Boston, MA
Bachelor of Science in Applied Mathematics *2026*

- **Relevant Coursework:** Robotics & Autonomous Systems, Linear Algebra, Probability, Stochastic Processes, Mathematical Statistics

Experience

Research Software Engineer, Robotics Jun 2023 – May 2026
Boston University College of Engineering *Boston, MA*

- Built parallel PyTorch/YOLO road-inspection pipeline processing 50K+ annotated frames across 64 km, achieving 94% classification accuracy; presented at Innovate@BU
- Architected SLAM-based reconstruction pipeline using PCL/Open3D, achieving sub-2 mm surface error with modular perception and mesh-optimization interfaces
- Built ROS 2 coordination layer for multi-drone fleet on NVIDIA Jetson, sustaining 500 GB+ daily sensor throughput and sub-50 ms inter-drone latency across 15+ containerized AWS deployments

Mobile Engineer Intern, iOS Jun 2025 – Aug 2025
Fidelity Investments, NetBenefits Health *Merrimack, NH*

- Refactored legacy monolithic iOS document-rendering engine into modular SwiftUI components, improving code maintainability and enabling 40% faster development cycles through declarative UI architecture
- Optimized content retrieval and rendering pipelines, achieving 70% latency reduction (300 to 90 ms) and improving user experience across 100K+ daily active sessions

Software Engineer, Backend & ML Jan 2024 – May 2024
Boston University Spark! Fellowship *Boston, MA*

- Architected and built ride-hailing ML backend from scratch in 4 months, serving ETA, pricing, and routing models through FastAPI and Docker on AWS, sustaining 150 req/s at 60 ms p95 in load testing
- Trained PyTorch ETA model on 100K historical trips, cutting mean arrival error 38% (6.2 to 3.8 min) vs. distance-based baseline on held-out test set, then reduced inference latency 67% (180 to 60 ms) with ONNX Runtime
- Delivered live MVP demo to 40+ stakeholders and partner investors, including DQN routing agent that cut simulated trip times 11% vs. shortest-path; the prototype was selected for 3-month pilot with 2 partners

Projects

Ring Dynamics | *Real-Time Pose-Based Combat Analytics System* | zhandolia.github.io/ring-dynamics

- Built YOLOv8-pose + ByteTrack pipeline that uses ring-zone limits and color-based re-identification to separate the 2 fighters from referees and spectators across camera cuts, running at about 24 FPS on CPU with no GPU
- Designed temporal scoring model that combines pose keypoints and motion into 4 EMA-smoothed metrics, producing estimated 10-9 round scores, strike events and per-second analytics timeline exported as structured JSON
- Built and deployed full-stack platform (FastAPI, Next.js) with background video upload and YouTube ingestion, RFC 7233 byte-range streaming, and dashboard that shows live metrics in sync with the annotated video

Zeta Engine | *Game-Theoretic Monte Carlo Poker Engine* | zhandolia.github.io/zeta-engine

- Engineered browser-based poker engine that runs 2,400 Monte Carlo hand evaluations per move in under 10 ms with no backend, deployed through GitHub Actions CI/CD pipeline
- Designed 8-term decision field recomputed across 6,400 points as a live heatmap each move; implemented Counterfactual Regret Minimization (CFR) in Python to solve Kuhn poker's Nash equilibrium

Technical Skills

Languages: Python, C++, Swift, Java, TypeScript, JavaScript, SQL
Robotics: ROS 2, SLAM, PCL, Open3D, NVIDIA Jetson, Linux
ML & Perception: PyTorch, YOLOv8, ONNX Runtime, ByteTrack, NumPy
Infrastructure: Docker, AWS, FastAPI, PostgreSQL, Git, CI/CD